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BLOCK 4 QUANTITATIVE METHODS-II

Many a times a social scientist or a policy maker is required to identify the factors causing variation in any social or economic issue of policy importance like food security, child deprivation etc. in order to take corrective policy measures. To handle such issues, factor analysis technique, in case of interval scale or ratio scale data and correspondence analysis method in nominal or categorized variables are used. Another Multi-Variate statistical model namely canonical correlation analysis is used to examine the interrelationships among set of multiple dependent variables and multiple independent variables. Cluster analysis enables us to classify the units or observations into sub-groups or clusters based on characteristics or information contained in the variables. Structural Equation Modeling (SEM) attempts to examine the causal relations by including directly observed variables and latent variables. Keeping in view the widening role of interdisciplinary nature of economics, the mixed methods research is gaining popularity among researchers. All these five Multi-Variate statistical techniques are increasingly being used in Mixed Methods Research and hence have been covered in this block.

Unit 13 on ‘**Multivariate Analysis: Factor Analysis**’ throws light on various concepts used in factor analysis, algorithm involved in factor analysis and the process to interpret the results of factor analysis.

Unit 14 on ‘**Canonical Correlation Analysis**’ deals with the procedure to compute the canonical correlation, its application in social sciences, interpretation of its results and limitation.

Unit 15 entitled ‘**Cluster Analysis**’ discusses the concept and purpose of cluster analysis, the various steps and algorithm in computation of its results and the different approaches followed in it.

Unit 16 on ‘**Correspondence Analysis**’ covers the concept and special features of correspondence analysis, its computation algorithm, interpretation of its results and the process of application of this technique in social sciences.

Unit 17 entitled ‘**Structural Equation Modeling (SEM)**’ throws light on the meaning and importance of SEM, its various models, steps involved in it, its advantages and disadvantages.

UNIT 13 MULTIVARIATE ANALYSIS: FACTOR ANALYSIS

Structure

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- 13.1 Introduction
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- 13.3 Historical Background of Factor Analysis
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13.0 OBJECTIVES

After going through this unit, you will be able to:

- explain the terms used in factor analysis model like factor loadings, specific variances, communalities, factor rotation and oblique rotation;
- know the various uses of factor analysis model;
- elucidate how to apply principle component analysis and maximum likelihood methods for estimating the parameters of a factor model;
- discuss methods for estimating factor score; and
- learn how to interpret the results of factor analysis based on rotated factor loadings, rotated eigenvalues and scree test.

13.1 INTRODUCTION

We have seen in Unit 12 that how the principal component analysis (PCA) is a useful technique to reduce the indicators or data for construction of composite index and identify the variables for policy decisions. Factor analysis is a further extension of principal component analysis. Many a times, a researcher in social sciences is confronted with entangled human behaviour, unknown interdependencies and masses of qualitative and quantitative variables. In such situations, he may intend to uncover major social, institutional and international pattern. Factor analysis enables him to manage over a hundred variables, explore a content area, structure a domain, map unknown concepts, classify or reduce data, illuminate causal nexus, define relationships, test hypotheses, formulate theories and make inferences. In short, a social scientist can discern the regularity and order in a phenomena under investigation by applying factor analysis. Factor analysis takes thousands of measurements and qualitative observations and resolves them into distinct patterns of occurrence. Factor analysis as a constituent of multi-variate analysis involves a large number of issues ranging from conceptual, application, modeling, inference and interpretation, factor rotation etc. Keeping in view, the limited scope of this unit, we shall confine here to explain the concepts used in factor analysis, methods of estimation in factor model and the techniques of extraction such as principal component method and maximum likelihood method. Let us begin with explaining the concept of factor analysis.

13.2 FACTOR ANALYSIS: CONCEPT AND MEANING

The term factor analysis refers to anyone of a number of similar but distinct multi-variate statistical models that model observed variables as linear functions of a set of latent or hypothetical variables (also known as factors) not directly observed.

Factor analysis models are similar to regression models in the sense that both types of models possess dependent variables as linear functions of independent variables. However, they are distinct to each other in the sense that in factor analysis models, the independent variables are not observed independently of the observed dependent variables.

The factor variables contained in factor analysis models may be determinate or indeterminate. The determinate models encompass the various component analysis models such as Principal Components Analysis, weighted principal components and Gultman's image analysis. Indeterminate models are represented by the common factor model that seeks to account for co-variation between the observed variables and a set of common factor variables. Determinate factor analysis are more useful in data reduction role by finding a smaller number of variables that capture most of the information of variation and co-variation among the observed variables. In contrast, the common factors in the indeterminate models are indeterminate from the observed variables and are not linear combinations of them. In this unit, we shall confine our discussion to determinate factor analysis.

Factor analysis is based on a model in which the observed vector is partitioned into an unobserved systematic part and an unobserved error part. The

components of the error vector are considered as uncorrelated or independent, while the systematic part is taken as a linear combination of a relatively small number of unobserved factor variables. The analysis separates the effects of the factors, which are of the basic interest from the errors. From another point of view the analysis gives a description or explanation of the interdependence of a set of variables in terms of the factors without regard to the observed variability.

13.3 HISTORICAL BACKGROUND OF FACTOR ANALYSIS

The early development of factor analysis was due to Charles Spearman in the year 1904. He published a paper that is thought to be where factor analysis originated. Spearman was working with examination scores of school children and he was able to notice certain systematic effects in the matrix of correlation between the scores the children made on different tests. For example, in one case he obtained the following matrix of correlations for school children in a preparatory school for their scores on tests in Classics (C), French (F), English (E), Mathematics (M), Discrimination of pitch (D), and Music (Mu) as follows:

	C	F	E	M	D	Mu
C	1.00	0.83	0.78	0.70	0.66	0.63
F	0.83	1.00	0.67	0.67	0.65	0.57
E	0.78	0.67	1.00	0.64	0.54	0.51
M	0.70	0.67	0.64	1.00	0.45	0.51
D	0.66	0.65	0.54	0.45	1.00	0.40
Mu	0.63	0.57	0.51	0.51	0.40	1.00

It can be noted out that this matrix has the interesting property that any two rows are almost proportional if the diagonals are ignored. Thus for rows C and E there are ratios:

$$\frac{0.83}{0.67} \approx \frac{0.70}{0.64} \approx \frac{0.66}{0.54} \approx \frac{0.63}{0.51} \approx 1.2$$

Spearman proposed the idea that the six test scores are all of the form

$$X_i = l_i F + \varepsilon_i$$

Where X_i is the i^{th} standardized score with a mean of zero and a standard deviation of one, l_i is a constant, F is a factor value which has mean zero and standard deviation one for individuals as a whole and ε_i is the part of X_i that is specific to the i^{th} test only. On the basis of his work Spearman formulated his two-factor theory of mental tests i.e each test result is made up of two parts, one that is common to all tests is general intelligence and another that is specific to the test. Later on this theory was modified to allow for each test result to consist of a part due to several common factors plus a part specific to test. This gives the general factor analysis model

$$X_i = l_{i1}F_1 + l_{i2}F_2 + \cdots + l_{im}F_m + \varepsilon_i$$

13.4 THE ORTHOGONAL FACTOR MODEL

In this section, first we will define some notations and terminology along with the underlying assumptions.

13.4.1 Notations and Terminology

Multivariate population and sample can be defined by mean vector μ and covariance matrix Σ . These are defined as follows. Suppose that there are p variables $X_1, X_2, \dots, X_p$ then the vector of sample mean is given by

$$\bar{X} = \begin{bmatrix} \bar{X}_1 \\ \bar{X}_2 \\ \vdots \\ \bar{X}_p \end{bmatrix}$$

This can be thought of as the centre of the sample. It is an unbiased estimate of the population vector of means

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix}$$

The matrix of the sample variance and covariances is

$$S = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1m} \\ S_{21} & S_{22} & \cdots & S_{2m} \\ \vdots & \vdots & & \vdots \\ S_{p1} & S_{p2} & & S_{pm} \end{bmatrix}$$

where S is called the sample covariance matrix or sometimes sample dispersion matrix. It is an unbiased estimate of the population covariance matrix Σ .

Consider m unobserved common factors $F_1, F_2, \dots, F_m$. The i^{th} common factor is F_i . Generally, m is substantially less than p .

The common factors are also formed into a vector as

$$F = \begin{bmatrix} F_1 \\ F_2 \\ \vdots \\ F_m \end{bmatrix}$$

13.4.2 The Factor Model

The factor model can be observed as a series of multiple regressions, predicting each of the observable variable X_i from the unobservable common factors F_i . In general the factor analysis model is given by

$$X_1 = \mu_1 + l_{11}F_1 + l_{12}F_2 + \cdots l_{1m}F_m + \varepsilon_1$$

$$X_2 = \mu_2 + l_{21}F_1 + l_{22}F_2 + \cdots l_{2m}F_m + \varepsilon_2$$

$$\vdots$$

$$X_p = \mu_p + l_{p1}F_1 + l_{p2}F_2 + \cdots l_{pm}F_m + \varepsilon_p$$

Here, the means $\mu_1, \mu_2, \dots, \mu_p$ can be regarded as the intercept for the multiple regression models. The regression coefficients l_{ij} (slopes) for all the multiple regressions are called factor loadings. Here, l_{ij} is the loading of the i^{th} variable on the j^{th} factor. These will be shown in the matrix form as follows:

$$\mathbf{L} = \begin{bmatrix} l_{11} & l_{12} & \cdots & l_{1m} \\ l_{21} & l_{22} & \cdots & l_{2m} \\ \vdots & \vdots & & \vdots \\ l_{p1} & l_{p2} & & l_{pm} \end{bmatrix}$$

and finally the errors ε_i are called the specific factors. Here, ε_i is the specific factor for variable i . The specific factors are also shown into a vector form as

Vector of specific factors

$$\varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_p \end{bmatrix}$$

In general, the basic factor analysis model is like a regression analysis model. Each of our response variable X is expressed as a linear function of the unobserved common factors $F_1, F_2, \dots, F_m$. Therefore, here one can say that we have m unobserved factors that control the variation among the data.

We will write the above factor model in matrix notation as

$$X = \mu + L F + \varepsilon \quad (13.1)$$

In general we want $m < p$

13.4.3 Model Assumptions

Some assumptions are necessary to uniquely estimate the parameters. Otherwise, an infinite number of equally well fitted models with different values for the parameters may be obtained if these assumptions are not made. Therefore, the following assumptions are made about the unobserved random vectors F and ε and its certain covariance relationships:

- i) The specific factors or random errors all have mean zero i.e.,

$$E(\varepsilon_i) = 0; \quad i = 1, 2, \dots, p$$

- ii) The common factors also have mean zero i.e.,

$$E(F_j) = 0; \quad j = 1, 2, \dots, m$$

- iii) The variance of specific factor i is Ψ_i , where Ψ_i is called the specific variance, and they are independent, i.e. $Cov(\varepsilon_i, \varepsilon_{i'}) = 0$ for $i \neq i'$.

$$Cov(\varepsilon_i) = E(\varepsilon_i, \varepsilon_i') = \Psi = \begin{bmatrix} \psi_1 & 0 & \cdots & 0 \\ 0 & \psi_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \psi_p \end{bmatrix}; \quad i = 1, 2, \dots, p$$

- iv) The common factors F_j have variance one and are uncorrelated, i.e.

$$V(F_j) = Cov(F_j, F_j') = 1 \text{ for } j = 1, 2, \dots, m$$

Thus co-variance matrix of F is an identity matrix I , i.e. $cov(F) = I$

- v) Since the common factors and specific factors are independent, then the specific factors are uncorrelated with the common factors i.e

$$\text{Cov}(\epsilon_i, F_j) = E(\epsilon_i, F_j') = 0; \quad i = 1, 2, \dots, p; \quad j = 1, 2, \dots, m$$

Now, the factor analysis model implies a covariance structure for the random variable X is given by equation

$$\begin{aligned} (X - \mu)(X - \mu) &= (L F + \epsilon)(L F + \epsilon) \\ &= (L F + \epsilon)((L F) + \epsilon) \\ &= L F(L F) + \epsilon(L F) + L F \epsilon + \epsilon \epsilon \end{aligned} \quad (13.1)$$

so that

$$\begin{aligned} \Sigma &= \text{Cov}(X) = E(X - \mu)(X - \mu) \\ &= L E(F F) L + E(\epsilon F) L + L E(F \epsilon) + E(\epsilon \epsilon) \\ \therefore \Sigma &= L L + \Psi \end{aligned} \quad (13.2)$$

This is the matrix of factor loadings, times its transpose plus a diagonal matrix containing the specific variances. Hence a simplified model for the covariance matrix is obtained which is used for estimation.

Also, on multiplying F in equation (13.1) on both sides, we have

$$\begin{aligned} (X - \mu)F &= (L F + \epsilon)F \\ &= L F F + \epsilon F \end{aligned}$$

so,

$$\text{Cov}(X, F) = E(X - \mu)F = L E(F F) + E(\epsilon F) = L$$

or

$$\text{Cov}(X_i, F_j) = l_{ij}$$

Under this model, the variance for the i^{th} observed variable is equal to the sum of the squared loadings for that variables and specific variance. The variance of variables X_i is:

$$\sigma_{ii} = \sum_{j=1}^m l_{ij}^2 + \psi_i \quad (13.3)$$

This is a derivation which is based on the previous assumptions and the quantity $\sum_{j=1}^m l_{ij}^2$ is called the communality for variable i .

Important Remarks:

- 1) The model $X - \mu = L F + \epsilon$ assumes that the data is a linear function of the common factors. However, since the common factors are not observable, we cannot check the linearity of the model.
- 2) The covariance matrix is a symmetric matrix, that is the variance between variables i and j is the same as the variance between j and i . For this model:

$$\Sigma = L L' + \Psi$$

The covariance matrix is going to have $p(p+1)/2$ unique elements of Σ which are approximated by mp factor loadings in the matrix L and the p specific variances ψ_i . When $m = p$, any covariance matrix Σ can be reproduced exactly as $L L'$, so Ψ can be the zero matrix. However, when m is smaller relative to p , that factor analysis is most useful. In this case,

the factor model provides a simple explanation of the covariation in X with fewer parameters than the $p(p+1)/2$ parameters in Σ .

For example, if X contains $p=12$ variables and the factor model with $m=2$ is appropriate, then the $p(p+1)/2 = 78$ elements of Σ are described in terms of the $mp + p = 36$ parameters l_{ij} and ψ_i of the factor model.

Unfortunately, for the factor analyst, most covariance matrices cannot be factored as $LL' + \Psi$, where the number of factors m is much less than p . There are some problems, where it is not possible to get a unique consistent solution for the parameters l_{ij} and ψ_i from the variances and covariances of the observable variables.

- 3) If $m > 1$, there is always some inherent ambiguity associated with the factor model. To explain that, let T be any $m \times m$ orthogonal matrix. A matrix is orthogonal if its inverse is equal to the transpose of the original matrix.

$$TT' = T'T = I$$

Hence, we can write the factor model in matrix notation as:

$$X - \mu = LF + \varepsilon = LTT'F + \varepsilon = L^*F^* + \varepsilon$$

Note that this does not change the calculations since the identity matrix times any matrix just gives back the original matrix. This results in an alternative factor model, where the relationship between the new factor loadings and the original factor loadings is:

$$L^* = LT$$

and the relationship between the new common factors and the original common factor is:

$$F^* = T'F$$

This gives a model that fits equally well. Moreover, since there are infinite numbers of orthogonal matrices, then there are an infinite number of alternative models. This model, as it turns out, satisfied all the assumptions that we discussed earlier as:

$$E(F^*) = E(T'F) = T'E(F) = 0$$

$$\text{Cov}(F^*) = \text{Cov}(T'F) = T'\text{Cov}(F)T = T'I T = T'T = I$$

and

$$\text{Cov}(F^*, \varepsilon) = \text{Cov}(T'F, \varepsilon) = T'\text{Cov}(F, \varepsilon) = T'0 = 0$$

On the basis of observations on X , it is impossible to distinguish the loadings L from the loadings L^* . That is, the factors F and $F^* = T'F$ have the same statistical properties, and even though the loadings L^* are, in general different from the loadings L , they both have the same covariance matrix Σ . That is

$$\Sigma = LL' + \Psi = LTT'L' + \Psi = (L^*)(L^*)' + \Psi \quad (13.4)$$

This ambiguity is going to be used to justify the factor rotation, since orthogonal matrices correspond to rotation of the coordinate system for X . We will use later to obtain more thrifty description of the data.